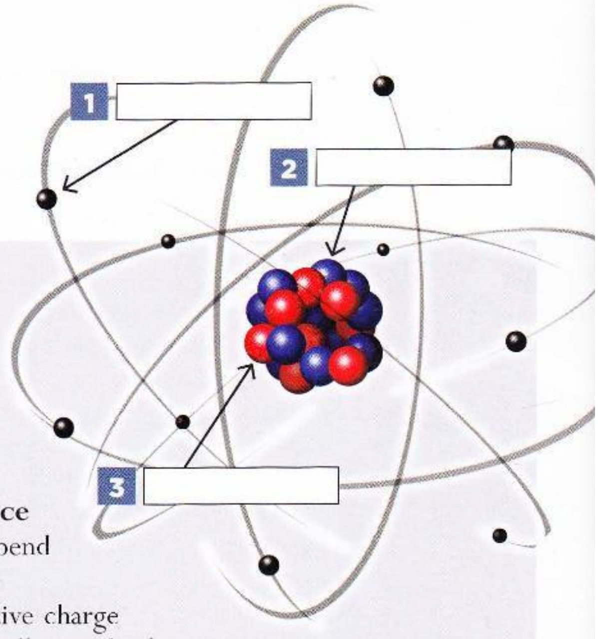
4 What is electricity?

1 **Read the text and label the picture with the name of each part.**



All substances, solids, liquids or gases, are composed of one or more of the chemical elements. Each element is composed of identical atoms.
Each atom is composed of a small central nucleus consisting of protons and neutrons around which **orbit shells** of electrons. These electrons are very much smaller than protons and neutrons.
The electrons in the **outermost** shell are called **valence** electrons and the electrical **properties** of the substance depend on the number of these electrons.
Neutrons have no electric **charge**, but protons have a positive charge while electrons have a negative charge. In some substances, usually metals, the valence electrons are free to move from one atom to another and this is what constitutes an electric current.

2 **Read the text again and complete the sentences with the missing information.**

1 Elements make up ______
2 Identical atoms ______
3 Atoms consist of ______, ______ and ______
4 Inside there are ______ and ______, while outside ______
5 Shells ______
6 Valence electrons ______
7 Neutrons do not have ______
8 Electricity is generated when ______

3 5 **Listen and complete the text with the missing information.**

Electricity consists of a (1) ______ of free electrons along a conductor. To produce this **current flow**, a generator is placed at the end of the conductor in order to move the (2) ______.

Conductors

Electricity needs a material which allows a current to **pass through** easily, which offers little (3) ______ to the flow and is full of free electrons. This material is called a conductor and can be in the form of a bar, tube or sheet. The most commonly used (4) ______ are wires, available in many sizes and **thicknesses**. They are **coated** with insulating materials such as plastic.

Semiconductors

Semiconductors such as silicon and germanium are used in transistors and their conductivity is **halfway** in between a conductor and an (5) ______.
Small quantities of other substances, called **impurities**, are introduced in the material to (6) ______ the conductivity.

Insulators

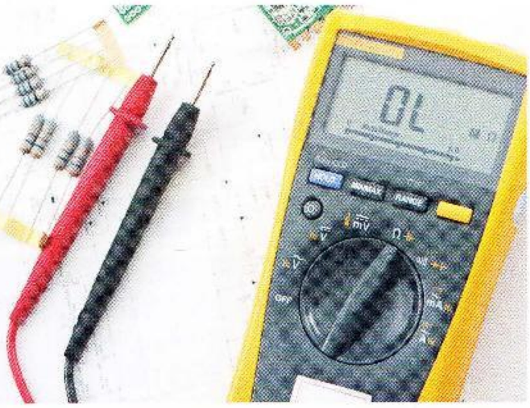
A material which contains very (7) ______ electrons is called an insulator. Glass, rubber, dry wood and (8) ______ resist the flow of electric charge, and as such they are good insulating materials.

4 Read the text again and decide if the following statements are true (T) or false (F), then correct the false ones.

1 A flow of electrons moving inside a conductor creates an electric current. ___
2 A generator is used to move the charges. ___
3 Electrons can easily pass through any material. ___
4 Any material is a good conductor. ___
5 Conductors are coated with insulators. ___
6 The presence of free electrons affects the conductivity of materials. ___
7 Impurities are introduced to increase conductivity. ___
8 Insulating materials resist the flow of electrons. ___

5 Read the text and complete the table with the missing information.

There are two types of current: Direct current (DC) and Alternating current (AC).
Direct current is a continuous flow of electrons in one direction and it never changes its direction until the power is stopped or **switched off**.
Alternating current constantly changes its direction because of the way it is generated. The term 'frequency' is used to indicate how many times the current changes its direction in one second.
Alternating current has a great advantage over direct current because it can be transmitted over very long distances through small wires, by making energy high voltage and low current.
There are several quantities that are important when we are talking about electric current. Volts (V) – so **named after** the Italian physicist Alessandro Volta – measure the difference of electric potential between two points on a conducting wire. Amperes (A) measure the amount of current flowing through a conductor, that is to say the number of electrons passing a point in a conductor in one second.



Coulomb (C) measure the quantity of charge transferred in one second by a **steady** current of one ampere. Power is the rate at which work is performed and it is measured in watts (W). A Kilowatt (kW), which is equal to one thousand watts, is used to measure the amount of used or available energy. The amount of electrical energy consumed in one hour at the constant rate of one kilowatt is called kilowatt-hour.

Unit of measurement	What does it measure?
(1) ______	the number of electrons passing a given point in a conductor in one second
(2) ______	the quantity of electricity transferred by a steady current of one ampere
(3) ______	the amount of electric energy used
(4) ______	the difference of potential between two points on a conductor
(5) ______	rate at which work is done

MY GLOSSARY

charge /tʃɑːdʒ/ ______
coated /kəʊtɪd/ ______
conductor /kən'dʌktə(r)/ ______
current flow /kʌrnt fləʊ/ ______
halfway /hɑːfweɪ/ ______
impurity /ɪm'pjʊərɪti/ ______
insulator /ɪnsjʊleɪtə(r)/ ______
to name after /tə neɪm 'ɑːftə(r)/ ______
to orbit /tuː 'ɔːbɪt/ ______
to pass through /tə pɑːs θruː/ ______
property /prɒpəti/ ______
semiconductor /semikən'dʌktə(r)/ ______
shell /ʃel/ ______
steady /stedi/ ______
to switch off /tə swɪtʃ ɒf/ ______
thickness /θɪknəs/ ______
valence /væləns/ ______